

Aneesh Srinivas

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EDUCATION

San Diego State University, San Diego, CA
M.S., Computational Science

Aug 2024 – Aug 2026

University of California, Berkeley, Berkeley, CA
B.A., Applied Mathematics & Computer Science

May 2020 – May 2024

Relevant Skills & Coursework: Artificial Intelligence & Machine Learning, Computer Vision and Image Processing, Software Development and Algorithmic optimization, Engineering Test Automation & Scripting, Open Source API Development, Numerical Analysis, Scientific Computing, Mathematical Modeling & Simulations, Real Analysis

SKILLS

Languages: Java, Python, Julia, C, C++, SQL, MATLAB, Mathematica

Tools/Libraries: NumPy, Pandas, Git, Matplotlib Animation, Pygame, scikit-image, SharePoint, Power Automate, Power Apps

EXPERIENCE

NRIEP Fall Intern

Oct 2025 – Dec 2025

Naval Research Enterprise Internship Program, NSWC Corona Division

- Worked with a multidisciplinary team of five students in the Readiness Assessment Division to design a secure workflow centralizing signature approvals in a single web tool and improving the user experience over email-based approval processes.
- Project scope was limited to workflow design and definition due to the 2025 U.S. Government shutdown.

Graduate Research & Teaching Assistant

Aug 2024 – May 2026

San Diego State University, Mathematics & Statistics Department

- Research computational techniques to improve and minimize grid folding, with applications in computer vision, including edge detection and deblurring, and machine learning.
- Supported grading, exam proctoring, tutoring, and office hours.

Summer Research Intern

May 2023 – Aug 2023

Emory University

- Analyzed environmental and socioeconomic data to predict social vulnerability maps; built an optimized model-driven framework and evaluated LSTM networks for NO₂ concentration forecasting, identifying strong SVI–NO₂ correlations in 2010. [Project | Poster](#)

GRADUATE THESIS & RESEARCH

Generating Ideal Reference Grid for Irregular Geometries

M.S. Thesis, Computational Science — San Diego State University

MATLAB | Numerical PDEs | Grid Generation | Optimization

- Developed a new computational method for generating reliable grids on complex and irregular shapes, addressing a common failure in traditional methods where grid lines overlap or “fold,” making downstream simulations unusable.
- **Impact** Eliminated folded cells across all primary test geometries and maintained valid results across 4 different grid resolutions, while the standard Laplace-based baseline failed on the same benchmark cases.
- Demonstrated that better grid quality can have a major computational impact: on a difficult Swan-shaped geometry, the improved grid reduced the required explicit simulation workload from approximately **679 million time steps to 292 thousand**—more than a **2,300× reduction**.
- Demonstrated method accuracy on a real-world simulation, showing approximately **second-order convergence** on a heat-equation benchmark and reducing numerical error by roughly **94%**.
- **Industry relevance:** My research can support applications that depend on accurate computation over irregular domains, including **AI/ML and scientific machine learning, computer vision and image processing**, engineering simulation, and **financial mathematics** models that rely on stable numerical PDE solvers.
- Implemented the workflow in MATLAB and designed it for integration with the open-source [MOLE grid-generation workflow](#).

SEMINARS AND INDUSTRY PRESENTATIONS

Higher-Order Mimetic Operators / ACSESS 2025 Presentation

Computational Science Research Center, San Diego State University

- Constructed higher-order mimetic operators using operator calculus as an alternative to traditional Vandermonde-matrix approaches, demonstrating **more than 400% performance improvement** while maintaining numerical stability.

Generating Ideal Reference Grid for Irregular Geometries / ACSESS 2026 Presentation

Computational Science Research Center, San Diego State University

- Presented a new approach to a common simulation problem: traditional grid-generation methods can produce overlapping or “folded” cells on irregular shapes, making the resulting computational model unreliable.

PROJECTS & OPEN-SOURCE WORK

MOLE (Mimetic Operators Library Enhanced)

[Repository](#) [Authors](#)

Co-authored and extended the library with additional functional APIs, improved documentation, and cross-platform compatibility. This is one of my many open source contributions.

Intelligent Search & Adversarial Game Agents

[GitHub](#)

Implemented A* and uniform-cost search for Wilson-algorithm mazes using a modified Manhattan metric; also built a Connect-4 agent using Minimax and alpha-beta pruning, achieving approximately 78% win efficiency across 300 random-user trials.

Thermal Hotspot Modeling on Printed Circuit Boards

[GitHub](#)

Developed and optimized a finite-element simulation incorporating electrical, physical, and thermal component properties to identify PCB thermal hotspots and potential failure points.

Efficient 2D/3D Mesh Generator for Arbitrary Polygons

[GitHub](#)

Designed a mesh generator supporting arbitrary polygons, multiple boundary conditions, edge intersections, and internal points; reduced mesh-generation time by approximately 40%.

Image Processing & Manufacturing Defect Detection

[GitHub](#)

Developed customized edge-detection and feature-detection methods for PCB manufacturing irregularities and object identification, including Sobel and Chan–Vese-based approaches.

Time-Series Forecasting with Kalman Filters

[GitHub](#)

Implemented a Kalman-filter model for short-term stock-price forecasting; identified overfitting under volatile conditions and began parameter tuning and regularization work to improve generalization.